MATH I – Week 7

Derivatives of elementary functions: derivatives of polynomial, rational, exponential, logarithmic, trigonometric functions.

[Continuation of previous week's (Week 6) topics]

Reading from textbook sections:

§3.5 Derivatives of Trigonometric Functions.

§3.9 Derivatives of Exponential and Logarithmic Functions.

Lecture prepared and presented by G. Neofotistos

Review of (selected) last week's material

Differentiation rules

Theorem 3.3

The Power Rule

Let n be a positive integer. If $f(x) = x^n$, then

$$f'(x) = nx^{n-1}.$$

Alternatively, we may express this rule as

$$\frac{d}{dx}x^n = nx^{n-1}.$$

Theorem 3.4

Sum, Difference, and Constant Multiple Rules

Let $f(x)$ and $g(x)$ be differentiable functions and k be a constant. Then each of the following equations holds.

Sum Rule. The derivative of the sum of a function f and a function g is the same as the sum of the derivative of f and the derivative of g.

$$\frac{d}{dx}(f(x) + g(x)) = \frac{d}{dx}(f(x)) + \frac{d}{dx}(g(x));$$

that is,

$$\text{for } j(x) = f(x) + g(x), j'(x) = f'(x) + g'(x).$$

Difference Rule. The derivative of the difference of a function f and a function g is the same as the difference of the derivative of f and the derivative of g:

$$\frac{d}{dx}(f(x) - g(x)) = \frac{d}{dx}(f(x)) - \frac{d}{dx}(g(x));$$

that is,

$$\text{for } j(x) = f(x) - g(x), j'(x) = f'(x) - g'(x).$$

Constant Multiple Rule. The derivative of a constant k multiplied by a function f is the same as the constant multiplied by the derivative:

$$\frac{d}{dx}(kf(x)) = k\frac{d}{dx}(f(x));$$

that is,

$$\text{for } j(x) = kf(x), j'(x) = kf'(x).$$

Theorem 3.5

Product Rule

Let $f(x)$ and $g(x)$ be differentiable functions. Then

$$\frac{d}{dx}(f(x)\,g(x)) = \frac{d}{dx}(f(x)) \cdot g(x) + \frac{d}{dx}(g(x)) \cdot f(x).$$

That is,

$$\text{if } j(x) = f(x)\,g(x), \text{ then } j'(x) = f'(x)\,g(x) + g'(x)\,f(x).$$

This means that the derivative of a product of two functions is the derivative of the first function times the second function plus the derivative of the second function times the first function.

In-Class activity

Applying the Product Rule to Binomials

For $j(x) = (x^2 + 2)(3x^3 - 5x)$, find $j'(x)$ by applying the product rule.

Theorem 3.5

Product Rule

Let $f(x)$ and $g(x)$ be differentiable functions. Then

$$\frac{d}{dx}(f(x)g(x)) = \frac{d}{dx}(f(x)) \cdot g(x) + \frac{d}{dx}(g(x)) \cdot f(x).$$

That is,

$$\text{if } j(x) = f(x)g(x), \text{ then } j'(x) = f'(x)g(x) + g'(x)f(x).$$

This means that the derivative of a product of two functions is the derivative of the first function times the second function plus the derivative of the second function times the first function.

In-Class activity

Applying the Product Rule to Binomials

For $j(x) = (x^2 + 2)(3x^3 - 5x)$, find $j'(x)$ by applying the product rule.

Solution

If we set $f(x) = x^2 + 2$ and $g(x) = 3x^3 - 5x$, then $f'(x) = 2x$ and $g'(x) = 9x^2 - 5$. Thus,

$$j'(x) = f'(x)g(x) + g'(x)f(x) = (2x)\left(3x^3 - 5x\right) + (9x^2 - 5)(x^2 + 2).$$

Simplifying, we have

$$j'(x) = 15x^4 + 3x^2 - 10.$$

In-Class activity

Theorem 3.6

The Quotient Rule

Let $f(x)$ and $g(x)$ be differentiable functions. Then

$$\frac{d}{dx}\left(\frac{f(x)}{g(x)}\right) = \frac{\frac{d}{dx}(f(x)) \cdot g(x) - \frac{d}{dx}(g(x)) \cdot f(x)}{(g(x))^2}.$$

That is,

$$\text{if } j(x) = \frac{f(x)}{g(x)}, \text{ then } j'(x) = \frac{f'(x)\,g(x) - g'(x)\,f(x)}{(g(x))^2}.$$

The proof of the **quotient rule** is very similar to the proof of the product rule, so it is omitted here. Instead, we apply this new rule for finding derivatives in the next example.

EXAMPLE 3.25

Applying the Quotient Rule

Use the quotient rule to find the derivative of $k(x) = \frac{5x^2}{4x+3}$.

In-Class activity Solution

Theorem 3.6

The Quotient Rule

Let $f(x)$ and $g(x)$ be differentiable functions. Then

$$\frac{d}{dx}\left(\frac{f(x)}{g(x)}\right) = \frac{\frac{d}{dx}(f(x)) \cdot g(x) - \frac{d}{dx}(g(x)) \cdot f(x)}{(g(x))^2}.$$

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The proof of the **quotient rule** is very similar to the proof of the product rule, so it is omitted here. Instead, we apply this new rule for finding derivatives in the next example.

EXAMPLE 3.25

Applying the Quotient Rule

Use the quotient rule to find the derivative of $k(x) = \frac{5x^2}{4x+3}$.

Solution

Let $f(x) = 5x^2$ and $g(x) = 4x + 3$. Thus, $f'(x) = 10x$ and $g'(x) = 4$. Substituting into the quotient rule, we have

$$k'(x) = \frac{f'(x)\,g(x) - g'(x)\,f(x)}{(g(x))^2} = \frac{10x(4x+3) - 4(5x^2)}{(4x+3)^2}.$$

Simplifying, we obtain

$$k'(x) = \frac{20x^2 + 30x}{(4x+3)^2}.$$

3.5 Derivatives of Trigonometric Functions

Learning Objectives

3.5.1 Find the derivatives of the sine and cosine function.

3.5.2 Find the derivatives of the standard trigonometric functions.

3.5.3 Calculate the higher-order derivatives of the sine and cosine.

Theorem 3.8

The Derivatives of sin x and cos x

The derivative of the sine function is the cosine and the derivative of the cosine function is the negative sine.

$$\frac{d}{dx}(\sin x) = \cos x \tag{3.11}$$

$$\frac{d}{dx}(\cos x) = -\sin x \tag{3.12}$$

Graph of *sinx* and *cosx*

Explain why the derivative of sinx is the cosx

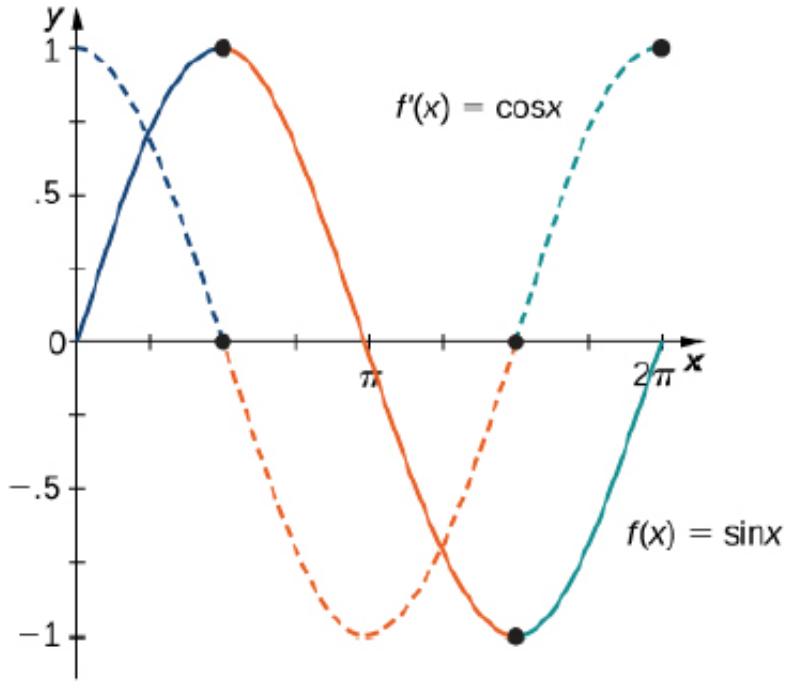


Figure 3.27 Where $f(x)$ has a maximum or a minimum, $f'(x) = 0$ that is, $f'(x) = 0$ where $f(x)$ has a horizontal tangent. These points are noted with dots on the graphs.

In-Class activity

Differentiating a Function Containing sin *x*

Find the derivative of $f(x) = 5x^3 \sin x$.

Differentiating a Function Containing sin *x*

Find the derivative of $f(x) = 5x^3 \sin x$.

In-Class activity
Solution
(Example 3.39 of textbook)

Solution

Using the product rule, we have

$$\begin{aligned} f'(x) &= \frac{d}{dx}\left(5x^3\right) \cdot \sin x + \frac{d}{dx}(\sin x) \cdot 5x^3 \\ &= 15x^2 \cdot \sin x + \cos x \cdot 5x^3. \end{aligned}$$

After simplifying, we obtain

$$f'(x) = 15x^2 \sin x + 5x^3 \cos x.$$

In-Class activity

Finding the Derivative of a Function Containing cos *x*

Find the derivative of $g(x) = \frac{\cos x}{4x^2}$.

In-Class activity Solution

Finding the Derivative of a Function Containing cos *x*

Find the derivative of $g(x) = \frac{\cos x}{4x^2}$.

Solution

By applying the quotient rule, we have

$$g'(x) = \frac{(-\sin x)4x^2 - 8x(\cos x)}{(4x^2)^2}.$$

Simplifying, we obtain

$$\begin{aligned} g'(x) &= \frac{-4x^2 \sin x - 8x \cos x}{16x^4} \\ &= \frac{-x \sin x - 2 \cos x}{4x^3}. \end{aligned}$$

In-Class activity

The Derivative of the Tangent Function

Find the derivative of $f(x) = \tan x$.

In-Class activity Solution

The Derivative of the Tangent Function

Find the derivative of $f(x) = \tan x$.

Solution

Start by expressing $\tan x$ as the quotient of $\sin x$ and $\cos x$:

$$f(x) = \tan x = \frac{\sin x}{\cos x}.$$

Now apply the quotient rule to obtain

$$f'(x) = \frac{\cos x \cos x - (-\sin x)\sin x}{(\cos x)^2}.$$

Simplifying, we obtain

$$f'(x) = \frac{\cos^2 x + \sin^2 x}{\cos^2 x}.$$

Recognizing that $\cos^2 x + \sin^2 x = 1$, by the Pythagorean theorem, we now have

$$f'(x) = \frac{1}{\cos^2 x}.$$

Finally, use the identity $\sec x = \frac{1}{\cos x}$ to obtain

$$f'(x) = \sec^2 x.$$

Do not memorize them!

Theorem 3.9

Derivatives of $\tan x$, $\cot x$, $\sec x$, and $\csc x$

The derivatives of the remaining trigonometric functions are as follows:

$$\frac{d}{dx}(\tan x) = \sec^2 x \tag{3.13}$$

$$\frac{d}{dx}(\cot x) = -\csc^2 x \tag{3.14}$$

$$\frac{d}{dx}(\sec x) = \sec x \tan x \tag{3.15}$$

$$\frac{d}{dx}(\csc x) = -\csc x \cot x. \tag{3.16}$$

In-Class activity

Finding Higher-Order Derivatives of $y = \sin x$

Find the first four derivatives of $y = \sin x$.

In-Class activity Solution

Finding Higher-Order Derivatives of $y = \sin x$

Find the first four derivatives of $y = \sin x$.

Solution

Each step in the chain is straightforward:

$$\begin{aligned} y &= \sin x \\ \frac{dy}{dx} &= \cos x \\ \frac{d^2y}{dx^2} &= -\sin x \\ \frac{d^3y}{dx^3} &= -\cos x \\ \frac{d^4y}{dx^4} &= \sin x. \end{aligned}$$

Analysis

Once we recognize the pattern of derivatives, we can find any higher-order derivative by determining the step in the pattern to which it corresponds. For example, every fourth derivative of sin *x* equals sin *x*, so

$$\frac{d^4}{dx^4}(\sin x) = \frac{d^8}{dx^8}(\sin x) = \frac{d^{12}}{dx^{12}}(\sin x) = \ldots = \frac{d^{4n}}{dx^{4n}}(\sin x) = \sin x$$

$$\frac{d^5}{dx^5}(\sin x) = \frac{d^9}{dx^9}(\sin x) = \frac{d^{13}}{dx^{13}}(\sin x) = \ldots = \frac{d^{4n+1}}{dx^{4n+1}}(\sin x) = \cos x.$$

In-Class activity

Using the Pattern for Higher-Order Derivatives of $y = \sin x$

Find $\frac{d^{74}}{dx^{74}}(\sin x)$.

In-Class activity Solution

Using the Pattern for Higher-Order Derivatives of $y = \sin x$

Find $\frac{d^{74}}{dx^{74}}(\sin x)$.

Solution

We can see right away that for the 74th derivative of $\sin x$, $74 = 4(18) + 2$, so

$$\frac{d^{74}}{dx^{74}}(\sin x) = \frac{d^{72+2}}{dx^{72+2}}(\sin x) = \frac{d^2}{dx^2}(\sin x) = -\sin x.$$

We will skip 3.9.3 for now.
We are going to cover it after the midterm
(…after we cover the *chain rule*)

3.9 Derivatives of Exponential and Logarithmic Functions

Learning Objectives

- 3.9.1 Find the derivative of exponential functions.
- 3.9.2 Find the derivative of logarithmic functions.
- 3.9.3 Use logarithmic differentiation to determine the derivative of a function.

Reminder of exponential and log properties

Rule: Laws of Exponents

For any constants $a > 0, b > 0$, and for all x and y,

1. $b^x \cdot b^y = b^{x+y}$
2. $\frac{b^x}{b^y} = b^{x-y}$
3. $(b^x)^y = b^{xy}$
4. $(ab)^x = a^x b^x$
5. $\frac{a^x}{b^x} = \left(\frac{a}{b}\right)^x$

Rule: Properties of Logarithms

If $a, b, c > 0, b \neq 1$, and r is any real number, then

1. $\log_b(ac) = \log_b(a) + \log_b(c)$ (Product property)
2. $\log_b\left(\frac{a}{c}\right) = \log_b(a) - \log_b(c)$ (Quotient property)
3. $\log_b(a^r) = r\log_b(a)$ (Power property)

$$e \approx 2.718282.$$

Theorem 3.14

Derivative of the Natural Exponential Function

Let $E(x) = e^x$ be the natural exponential function. Then

$$E'(x) = e^x.$$

In general,

$$\frac{d}{dx}\left(e^{g(x)}\right) = e^{g(x)}g'(x).$$

In-Class activity

Find the (first) derivative of the function:

a. $f(x) = e^{2x}$

b. $g(x) = e^{3x^2}$

c. $h(x) = e^{3x^2+5x^3}$

In-Class activity
Solution

Find the (first) derivative of the function:

a. $f(x) = e^{2x}$

$$\boxed{f'(x) = 2e^{2x}.}$$

b. $g(x) = e^{3x^2}$

$$\boxed{g'(x) = 6xe^{3x^2}.}$$

c. $h(x) = e^{3x^2+5x^3}$

$$\boxed{h'(x) = (6x + 15x^2)e^{3x^2+5x^3}.}$$

In-Class activity

Explain the graph

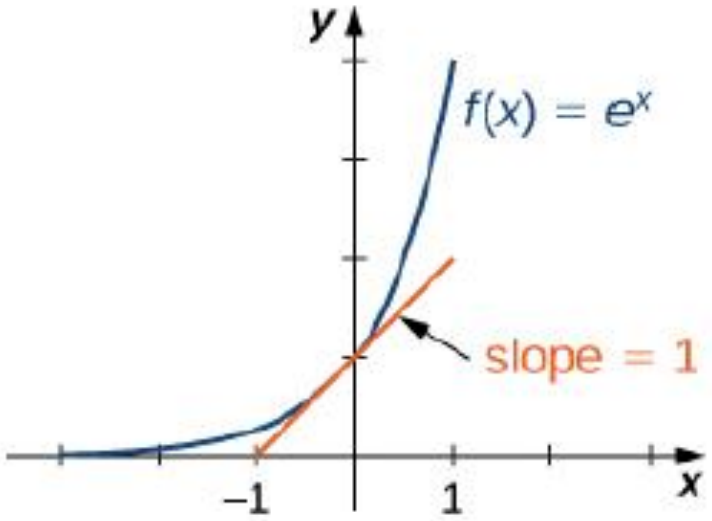


Figure 1.45 The graph of $f(x) = e^x$ has a tangent line with slope 1 at $x = 0$.

Example

EXAMPLE 3.76

Applying the Natural Exponential Function

A colony of mosquitoes has an initial population of 1000. After t days, the population is given by $A(t) = 1000e^{0.3t}$. Show that the ratio of the rate of change of the population, $A'(t)$, to the population, $A(t)$ is constant.

Solution

First find $A'(t)$. By using the chain rule, we have $A'(t) = 300e^{0.3t}$. Thus, the ratio of the rate of change of the population to the population is given by

$$\frac{A'(t)}{A(t)} = \frac{300e^{0.3t}}{1000e^{0.3t}} = 0.3.$$

The ratio of the rate of change of the population to the population is the constant 0.3.

Example

For

$$f(x) = 2^x,$$

the derivative is

$$\boxed{f'(x) = 2^x \ln 2.}$$

Step 1: Rewrite using the natural exponential

Since $2 = e^{\ln 2}$,

$$2^x = \left(e^{\ln 2}\right)^x = e^{x \ln 2}.$$

Thus,

$$f(x) = e^{x \ln 2}.$$

Using

$$\frac{d}{dx} e^{u(x)} = e^{u(x)} u'(x),$$

we obtain

$$f'(x) = e^{x \ln 2} \ln 2.$$

Substitute $e^{x \ln 2} = 2^x$:

$$\boxed{\frac{d}{dx} 2^x = 2^x \ln 2.}$$

Reminder

Optional

The exponential function $f(x) = b^x$ is one-to-one, with domain $(-\infty, \infty)$ and range $\left(0, \infty\right)$. Therefore, it has an inverse function, called the *logarithmic function with base b*. For any $b > 0, b \neq 1$, the logarithmic function with base b, denoted $\log_b$, has domain $(0, \infty)$ and range $\left(-\infty, \infty\right)$, and satisfies

$$\log_b(x) = y \text{ if and only if } b^y = x.$$

For example,

$$\begin{aligned}
&\log_2(8) = 3 && \text{since} && 2^3 = 8,\\
&\log_{10}\left(\tfrac{1}{100}\right) = -2 && \text{since} && 10^{-2} = \tfrac{1}{10^2} = \tfrac{1}{100},\\
&\log_b(1) = 0 && \text{since} && b^0 = 1 \text{ for any base } b > 0.
\end{aligned}$$

Furthermore, since $y = \log_b(x)$ and $y = b^x$ are inverse functions,

$$\log_b\left(b^x\right) = x \text{ and } b^{\log_b(x)} = x.$$

The most commonly used logarithmic function is the function $\log_e$. Since this function uses natural e as its base, it is called the **natural logarithm**. Here we use the notation $\ln(x)$ or $\ln x$ to mean $\log_e(x)$. For example,

$$\ln(e) = \log_e(e) = 1, \ln\left(e^3\right) = \log_e\left(e^3\right) = 3, \ln(1) = \log_e(1) = 0.$$

Since the functions $f(x) = e^x$ and $g(x) = \ln(x)$ are inverses of each other,

$$\ln\left(e^x\right) = x \text{ and } e^{\ln x} = x,$$

Rule: Change-of-Base Formulas

Let $a > 0, b > 0$, and $a \neq 1, b \neq 1$.

1. $a^x = b^{x\log_b a}$ for any real number x.
   If $b = e$, this equation reduces to $a^x = e^{x\log_e a} = e^{x\ln a}$.

2. $\log_a x = \dfrac{\log_b x}{\log_b a}$ for any real number $x > 0$.
   If $b = e$, this equation reduces to $\log_a x = \dfrac{\ln x}{\ln a}$.

Derivative of the Logarithmic Function

Now that we have the derivative of the natural exponential function, we can use implicit differentiation to find the derivative of its inverse, the natural logarithmic function.

Theorem 3.15

The Derivative of the Natural Logarithmic Function

If $x > 0$ and $y = \ln x$, then

$$\frac{dy}{dx} = \frac{1}{x}. \tag{3.30}$$

More generally, let $g(x)$ be a differentiable function. For all values of x for which $g(x) > 0$, the derivative of $h(x) = \ln(g(x))$ is given by

$$h'(x) = \frac{1}{g(x)} g'(x). \tag{3.31}$$

In-Class activity

Taking a Derivative of a Natural Logarithm

Find the derivative of $f(x) = \ln\left(x^3 + 3x - 4\right)$.

In-Class activity Solution

EXAMPLE 3.77

Taking a Derivative of a Natural Logarithm

Find the derivative of $f(x) = \ln\left(x^3 + 3x - 4\right)$.

Solution

Use Equation 3.31 directly.

$$f'(x) = \frac{1}{x^3+3x-4} \cdot \left(3x^2 + 3\right) \qquad \text{Use } g(x) = x^3 + 3x - 4 \text{ in } h'(x) = \frac{1}{g(x)} g'(x).$$

$$= \frac{3x^2+3}{x^3+3x-4} \qquad \text{Rewrite.}$$

In-Class activity

Using Properties of Logarithms in a Derivative

Find the derivative of $f(x) = \ln\left(\frac{x^2 \sin x}{2x+1}\right)$.

In-Class activity Solution

EXAMPLE 3.78

Using Properties of Logarithms in a Derivative

Find the derivative of $f(x) = \ln\left(\frac{x^2 \sin x}{2x+1}\right)$.

Solution

At first glance, taking this derivative appears rather complicated. However, by using the properties of logarithms prior to finding the derivative, we can make the problem much simpler.

$$f(x) = \ln\left(\frac{x^2 \sin x}{2x+1}\right) = 2\ln x + \ln(\sin x) - \ln(2x+1) \qquad \text{Apply properties of logarithms.}$$

$$f'(x) = \frac{2}{x} + \cot x - \frac{2}{2x+1} \qquad \text{Apply sum rule and } h'(x) = \frac{1}{g(x)} g'(x).$$

Real-world problem to solve

204. **[T]** The amount of rainfall per month in Phoenix, Arizona, can be approximated by $y(t) = 0.5 + 0.3\cos t$, where t is months since January. Find y' and use a calculator to determine the intervals where the amount of rain falling is decreasing.

The rainfall model is

$$y(t) = 0.5 + 0.3\cos t,$$

Solution

where t is the number of months since January. We need to find $y'(t)$ and determine **when the modeled monthly rainfall is decreasing**.

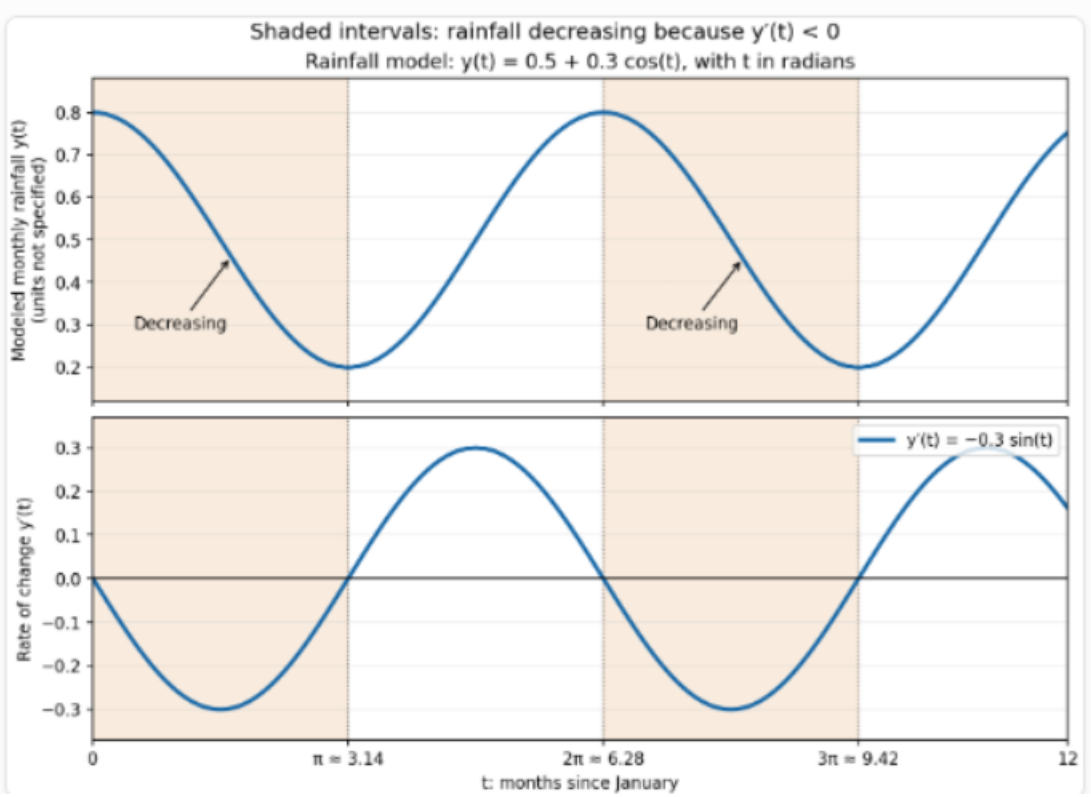
1. Differentiate

Using $\frac{d}{dt}\cos t = -\sin t$,

$$\boxed{y'(t) = -0.3\sin t.}$$

Use **radian mode** on your calculator.

Interval	Sign of $\sin t$	Sign of $y'(t)$	Rainfall
$(0, \pi)$	$+$	$-$	Decreasing
$(\pi, 2\pi)$	$-$	$+$	Increasing
$(2\pi, 3\pi)$	$+$	$-$	Decreasing
$(3\pi, 12)$	$-$	$+$	Increasing



Real-world problem

[T] The number of cases of influenza in New York City from the beginning of 1960 to the beginning of 1964 is modeled by the function $N(t) = 5.3e^{0.093t^2 - 0.87t}, (0 \le t \le 4)$, where $N(t)$ gives the number of cases (in thousands) and t is measured in years, with $t = 0$ corresponding to the beginning of 1960.

a. Show work that evaluates $N(0)$ and $N(4)$. Briefly describe what these values indicate about the disease in New York City.

b. Show work that evaluates $N'(0)$ and $N'(3)$. Briefly describe what these values indicate about the disease in New York City.

Solution:

The model is

$$N(t) = 5.3e^{0.093t^2 - 0.87t}, \qquad 0 \le t \le 4,$$

where $N(t)$ **is measured in thousands of cases** and t is measured in years from the beginning of 1960.

Thus, $t = 3$ corresponds to the beginning of 1963, and $t = 4$ to the beginning of 1964.

a. Evaluate and interpret $N(0)$ and $N(4)$

At $t = 0$,

$$N(0) = 5.3e^{0.093(0)^2 - 0.87(0)} = 5.3e^0 = \boxed{5.3}.$$

Since the units are thousands of cases, the model predicts

$$\boxed{5,300 \text{ influenza cases at the beginning of 1960.}}$$

At $t = 4$, first evaluate the exponent:

At $t = 4$, first evaluate the exponent:

$$0.093(4)^2 - 0.87(4) = 0.093(16) - 3.48 = 1.488 - 3.48 = -1.992.$$

Therefore,

$$N(4) = 5.3e^{-1.992} \approx \boxed{0.723}.$$

The model predicts approximately

$$\boxed{723 \text{ influenza cases at the beginning of 1964.}}$$

These values indicate that the modeled number of cases decreased substantially between the beginning of 1960 and the beginning of 1964.

b. Evaluate and interpret $N'(0)$ and $N'(3)$

First differentiate using the **chain rule**. Let

$$u(t) = 0.093t^2 - 0.87t.$$

Then

$$u'(t) = 0.186t - 0.87.$$

Since $\frac{d}{dt}e^{u(t)} = e^{u(t)}u'(t)$,

$$\boxed{N'(t) = 5.3e^{0.093t^2 - 0.87t}(0.186t - 0.87).}$$

Equivalently,

$$N'(t) = N(t)(0.186t - 0.87).$$

The derivative has units of **thousands of cases per year**.

At $t = 0$,

$$N'(0) = 5.3e^0(-0.87) = \boxed{-4.611}.$$

Thus, at the beginning of 1960, the modeled number of influenza cases was **decreasing at an instantaneous rate of**

$$\boxed{4,611 \text{ cases per year.}}$$

At $t = 3$,

$$0.093(3)^2 - 0.87(3) = 0.837 - 2.61 = -1.773,$$

and

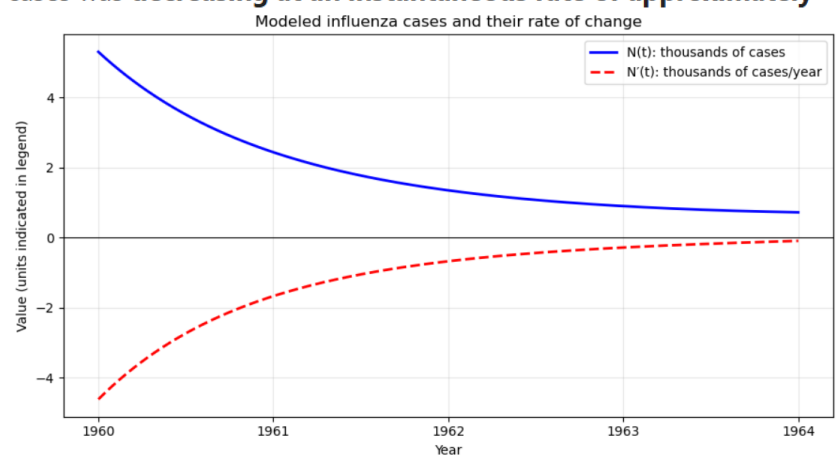
$$0.186(3) - 0.87 = 0.558 - 0.87 = -0.312.$$

Consequently,

$$N'(3) = 5.3e^{-1.773}(-0.312) \approx \boxed{-0.280819}.$$

Therefore, at the beginning of 1963, the modeled number of cases was **decreasing at an instantaneous rate of approximately**

$$\boxed{281 \text{ cases per year.}}$$



Quantity	Model value	Interpretation
$N(0)$	5.3 thousand cases	Approximately 5,300 cases at the beginning of 1960
$N(4)$	0.723 thousand cases	Approximately 723 cases at the beginning of 1964
$N'(0)$	-4.611 thousand cases/year	Cases decreasing at about 4,611 cases/year initially
$N'(3)$	-0.280819 thousand cases/year	Cases decreasing at about 281 cases/year at the beginning of 1963

Interpretation of the derivatives: both are negative, indicating declining case numbers. The smaller magnitude in 1963 means the decline was slower then. These are **instantaneous rates**, not the total numbers of cases lost over the following year.